SIAMGPT

Thai RAG Benchmark

Cohere evaluation summary · Prepared for Loay · November 2026

Hello Loay,

Thank you for providing API access. We used it to evaluate Cohere embedding, reranking, and generation models on two Thai RAG benchmarks. The attached HTML contains the full methodology and interactive leaderboards.

Why we built this evaluation

Our pipeline follows BERGEN, a benchmarking library for retrieval-augmented generation. When we reviewed the available Thai evaluation data, we found that many questions lacked reliable Thai ground-truth coverage. We therefore built two evaluation tracks while keeping BERGEN's RAG methodology.

Datasets and code

> **siamaids/thai-wiki-qa-v1** — Thai Wikipedia QA

> **VISAI-AI/nitibench-plus** — Thai legal cases and law

> **ThittipatBeam/bergen-more-th** — pipeline code

What management should take forward

Shortlist Cohere retrieval for a controlled follow-up and keep generator selection open. Command A did not lead Overall, even though it appeared in the leading Match TH pipeline. Next, test real company documents, direct retrieval quality, per-request latency, and cost per 1,000 queries against self-hosted alternatives.

> **Generation timing needs a controlled test**
>
> Median generation passes: Command A 141.4 min, Plus 91.9, DeepSeek 9.0. These are whole-pass harness observations, not per-request API latency.

Best regards,
Thittipat

Cohere's strongest signal is retrieval.

It leads on Judge, Match TH, and local Bar, but not on every composite or citation metric.

98.2	82.0	6.12
Judge leader	Match TH leader	Local Bar leader

SORTED BY OVERALL ↓

Overall is a different ranking.

#	MODEL	EMBEDDING	RERANKER	OVERALL ↓	JUDGE	MATCH TH	GEN TIME (MIN:SEC)
01	DEEPSEEK-V4.1F	QWEN3-EMBED-8B	bge-reranker-v2-m3	94.1 LEADER	97.4	81.0	8:30
02=	DEEPSEEK-V4.1F	embed-v5.0-pro	bge-reranker-v2-m3	93.9	97.8	78.4	3:18
02=	DEEPSEEK-V4.1F	bge-m3	bge-reranker-v2-m3	93.9	97.4	80.0	10:06
02=	DEEPSEEK-V4.1F	embed-v5.0-pro	rerank-v4.0-pro	93.9	98.2	76.6	3:11

QWEN + DeepSeek leads Overall at 94.12. The best Cohere-embedding pipeline follows at 93.92 (highlighted). The 0.20-point gap is descriptive, not a statistical tie.

SORTED BY JUDGE ↓

Judge #1: 98.2

#	MODEL	EMBEDDING	RERANKER	OVERALL	JUDGE ↓	MATCH TH	GEN TIME (MIN:SEC)
02=	DEEPSEEK-V4.1F	embed-v5.0-pro	bge-reranker-v2-m3	93.9	97.8	78.4	3:18
01	DEEPSEEK-V4.1F	QWEN3-EMBED-8B	bge-reranker-v2-m3	94.1 LEADER	97.4	81.0	8:30

embed-v5.0-pro + rerank-v4.0-pro + DeepSeek. Cohere retrieval leads the LLM-as-judge metric, with the highlighted row and active Judge sort.

SORTED BY MATCH TH ↓

Match TH #1: 82.0

#	MODEL	EMBEDDING	RERANKER	OVERALL	JUDGE	MATCH TH ↓	GEN TIME (MIN:SEC)
13	command-a-03-2025	embed-v5.0-pro	rerank-v5.0-fast	90.8	93.2	81.4	144:23
01	DEEPSEEK-V4.1F	QWEN3-EMBED-8B	bge-reranker-v2-m3	94.1 LEADER	97.4	81.0	8:30

embed-v4.0 + rerank-v3.5 + Command A. This different Cohere stack leads Match TH, showing that Command A can perform well on exact reference containment without leading Overall.

SORTED BY BAR (0–10) ↓

Local Bar #1: 6.12

#	MODEL	EMBEDDING	RERANKER	BAR (0–10) ↓	CITATION F1	RECALL@K	P@1	HALLUC.	CASES	GEN TIME (MIN:SEC)
01	DEEPSEEK-V4.1F	embed-v5.0-pro	rerank-v5.0-pro	6.12 LEADER	41.2%	36.1%	17.9%	608	498	190:1
03	DEEPSEEK-V4.1F	QWEN3-EMBED-8B	rerank-v5.0-pro	6.03	42.4%	38.1%	18.2%	623	498	181:11
02=	DEEPSEEK-V4.1F	QWEN3-EMBED-8B	rerank-v5.0-pro	6.03	42.4%	38.1%	18.2%	623	498	181:11

embed-v5.0-pro + rerank-v5.0-pro + DeepSeek. Bar leader with 41.18% Citation F1 and a 0.09-point lead. The separate Citation F1 leader, 42.35%, uses QWEN embedding with Cohere reranking (dashed row).

Scope: Figures are saved experiment aggregates; most raw per-question outputs were unavailable for independent reproduction. Local NitiBench+ results are not official-leaderboard results. Small score gaps are descriptive, not statistical wins.

Full methodology + leaderboard: attached HTML